University of Ottawa

Department of Mathematics and Statistics

MAT 1302B: Mathematical Methods II

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Final Exam – Solutions

April 2017

Surname _____ First Name _____

Student # _____ Seat # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) This exam consists of two parts. Questions 1 through 14 are *answer only*. For these questions, only your final answer will be considered for marks. Questions 15 through 21 are *long answer*. For these questions, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

(c) The number of points available for each question is indicated in square brackets.

(d) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this *clearly*. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(e) Write your student number at the top of each page in the space provided.

(f) You may use the second-to-last page of the exam as extra space for solutions. If you do so, indicate clearly on the page of the relevant question that you have continued your solution on the extra page.

(g) You may use the last page of the exam as scrap paper.

(h) Cellular phones, unauthorized electronic devices or course notes are not allowed during this exam. Phones and devices must be turned off and put away in your bag. Do not keep them in your possession, such as in your pockets. If caught with such a device or document, the following may occur: you will be asked to leave immediately the exam and academic fraud allegations will be filed which may result in you obtaining a 0 (zero) for the exam.

By signing below, you acknowledge that you have ensured that you are complying with statement (h) above:

Signature: _____

Good luck!

Please do not write in the table below.

Question	1–14	15	16	17	18	19	20	21	Total
Maximum	27	5	4	8	6	3	4	3	60
Grade									

Part A: Answer Only Questions

For Questions 1–14, only your final answer will be considered for marks. If applicable, write your final answers in the spaces provided.

1. **[2 points]** Consider the matrices

$$A = \begin{bmatrix} 2 & -3 \\ 1 & -1 \\ 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ -1 & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}, \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Compute $-4I_2 + A^T B$.

Answer: $-4I_2 + A^T B = \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ -1 & -3 \end{bmatrix}$

2. **[2 points]** Let $z = 2i + 3$ and $w = 1 - i$. Write the complex number $\dfrac{-2}{z - \overline{w}}$ in the form $a + bi$ where a and b are real numbers.

Answer:

$$\frac{-2}{z - \bar{w}} = \frac{-2}{(3-1) + (2i - i)} = \frac{-2}{2 + i} = \frac{(-2)(2 - i)}{(2 + i)(2 - i)} = -\frac{4}{5} + \frac{2}{5}i.$$

3. **[2 points]** Let A, B, and C be 4×4 matrices such that $\det A = 2$, $\det B = -\dfrac{1}{3}$, and $\det(C) = 6$. Calculate $\det(-2AB^T CA^{-3})$.

Answer: $\det(-2AB^T CA^{-3}) = -8$.

4. **[2 points]** Determine all values of $t \in \mathbb{R}$ such that the linear system

$$\begin{cases} 2y_1 - 3y_2 + \frac{1}{2}y_3 = -2 \\ 4y_1 - 6y_2 + 2ty_3 = -3 \end{cases}$$

is consistent.

Answer: $t \neq \frac{1}{2}$

5. **[2 points]** Let A and B be two $m \times n$ matrices such that $n > m \geq 1$. Each of the statements below is either always true or always false. For each statement below, write '**T**' if the statement is always true, and write '**F**' if the statement is always false. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

F $\mathrm{Col}(A^T)$ is a subspace of $\mathbb{R}^m$.

T $\mathrm{Nul}(B) \neq \{0\}$

T $\det(AB^T) = \det(BA^T)$

F $\mathrm{rank}(AB^T) + \dim \mathrm{Nul}\,(AB^T) = n$.

6. **[2 points]** Let $A = \begin{bmatrix} 1+i & 0 & 0 \\ -2 & 1 - \frac{1}{i} & 0 \\ 1+2i & -i & -1 \end{bmatrix}$. Write down the eigenvalues of A and their multiplicities. You should write the eigenvalues in the form $a + bi$, where $a, b \in \mathbb{R}$.

Answer: Note that $1 - \frac{1}{i} = 1 + i$. Therefore $1 + i$ has multiplicity 2 and -1 has multiplicity 1.

7. **[1 point]** Suppose that A is a 6×6 matrix which has 4 non-pivot columns. Write down the dimension of the subspace $W = \{\mathbf{v} \in \mathbb{R}^6 \ : \ \text{the equation } A\mathbf{x} = \mathbf{v} \text{ has a solution } \mathbf{x} \in \mathbb{R}^6\}$.

Answer: W is equal to $\mathrm{Col}\, A$. Therefore the dimension of W is equal to the number of pivot columns, i.e., $6 - 4 = 2$.

8. **[2 points]** For each of the following subsets of $\mathbb{R}^3$, write 'Y' if the set is a subspace of $\mathbb{R}^3$, and write 'N' if it is not. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

Y $\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; -3x_1 + x_2 - 2x_3 = 0 \right\}$

N $\left\{ \begin{bmatrix} x+y \\ x-y \\ 2x-y \end{bmatrix} \;\middle|\; x, y \geq 0 \right\}$

Y The line in $\mathbb{R}^3$ that passes through the points $(1,0,0)$ and $(0,0,0)$.

Y $\left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; \begin{bmatrix} -2 & 3 & -2 \\ 0 & -2 & 2 \\ -\frac{1}{2} & -\frac{1}{2} & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2x \\ -2y \\ -2z \end{bmatrix} \right\}$

9. **[1 point]** Suppose that A and B are 4×7 matrices, and suppose that $C = A + \frac{1}{2}B$. Let $\mathbf{x}$ be a vector in $\mathbb{R}^7$ such that

$$A\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 1 \\ 4 \end{bmatrix} \quad \text{and} \quad B\mathbf{x} = \begin{bmatrix} 4 \\ 2 \\ -6 \\ 4 \end{bmatrix}.$$

Calculate $C\mathbf{x}$.

Solution: $C\mathbf{x} = A\mathbf{x} + \frac{1}{2}B\mathbf{x} = \begin{bmatrix} 3 \\ -1 \\ -2 \\ 6 \end{bmatrix}$

10. **[3 points]** Let A be an $n \times n$ matrix, and consider the sentence

"A is invertible." $(\star)$

For each statement below, write 'Y' if it is equivalent to the sentence $(\star)$, and write 'N' if it is not equivalent to $(\star)$. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

Y $\det(A) \neq 0$.

N The homogeneous linear system $A\mathbf{x} = \mathbf{0}$ is consistent.

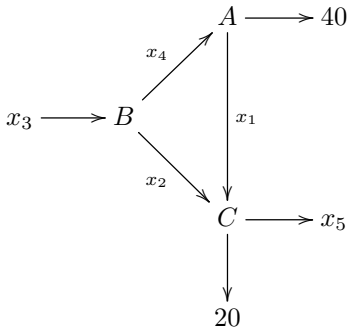
Y Every column vector in $\mathbb{R}^n$ can be expressed as a linear combination of the columns of A.

Y The columns of A form a basis for $\mathbb{R}^n$.

N The reduced echelon form of A has non-pivot columns.

N $\operatorname{rank} A = 0$.

11. **[2 points]** Consider the traffic flow described by the following diagram. The letters A through C label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 5$. (Do not solve the linear system.)

Solution:

$$\begin{cases} A: & x_4 = x_1 + 40 \\ B: & x_3 = x_2 + x_4 \\ C: & x_1 + x_2 = x_5 + 20 \\ \text{Total}: & x_3 = 60 + x_5 \end{cases}$$

12. **[2 points]** Let A be an 8×8 matrix such that the characteristic equation of A is

$$\lambda^8 - 3\lambda^3 + 2\lambda^2 = 0.$$

Also recall that I_8 denotes the 8×8 identity matrix. For each statement below, write 'T' if the statement is true, and write 'F' if the statement is false. You will receive 0.5 points for each correct answer, lose 0.25 points for each incorrect answer, and receive zero points for an answer left blank. You cannot receive a negative score on this question.

F A is invertible.

T There exists a nonzero vector $\mathbf{x} \in \mathbb{R}^8$ such that $A\mathbf{x} = \mathbf{x}$.

T The eigenvalue $\lambda = 0$ of A has multiplicity 2.

F The homogeneous linear system $(A - 2I_8)\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

13. **[2 points]** Write down $\det(A^3)$, where

$$A = \begin{bmatrix} 1 & 2 & -\frac{3}{5} & 0 \\ 0 & 2 & -1 & -7 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}.$$

Answer: -8.

14. **[2 points]** Let A and B be $n \times n$ invertible matrices. Solve the matrix equation $AB^TXB^2 + BA^T = 0$ for the matrix X.

Answer: $X = -(B^T)^{-1}A^{-1}BA^TB^{-2}$.

Part B: Long Answer Questions

For Questions 15–21, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

15. (a) **[4 points]** Is the following linear system consistent or inconsistent? If it is consistent, then write down the general solution in vector parametric form.

$$\begin{cases} x_1 + 2x_3 + 9x_4 &= 4x_2 + 1 \\ x_3 + 5x_4 &= -1 \\ 8x_2 - 2x_3 &= 2x_1 + 8x_4 - 4 \end{cases}$$

Solution: We reduce the augmented matrix of the system to the REF:

$$\left[\begin{array}{cccc|c} 1 & -4 & 2 & 9 & 1 \\ 0 & 0 & 1 & 5 & -1 \\ -2 & 8 & -2 & -8 & -4 \end{array}\right] \xrightarrow{R_3 \to R_3 + 2R_1} \left[\begin{array}{cccc|c} 1 & -4 & 2 & 9 & 1 \\ 0 & 0 & 1 & 5 & -1 \\ 0 & 0 & 2 & 10 & -2 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & -4 & 2 & 9 & 1 \\ 0 & 0 & 1 & 5 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & -4 & 0 & -1 & 3 \\ 0 & 0 & 1 & 5 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

It follows that the system is consistent (since the rightmost column is not pivot). Moreover, x_1, x_3 are basic, x_2, x_4 are free. The general solution is

$$\begin{cases} x_1 = 4x_2 + x_4 + 3 \\ x_2 = \text{free} \\ x_3 = -5x_4 - 1 \\ x_4 = \text{free} \end{cases}$$

The vector parametric form of the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4x_2 + x_4 + 3 \\ x_2 \\ -5x_4 - 1 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -5 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

(b) **[1 point]** Using your solution to part (a), write down the general solution of the following linear system in vector parametric form.

$$\begin{bmatrix} 1 & -4 & 2 & 9 \\ 0 & 0 & 1 & 5 \\ -2 & 8 & -2 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hint: Compare the augemented matrices of the two systems.

Solution: The second system is the homogeneous version of the previous system. Therefore its general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -5 \\ 1 \end{bmatrix}$$

16. [**4 points**] Calculate the determinant of the following matrix using the method of cofactor expansion.

$$M = \begin{bmatrix} 1 & 0 & -1 & 2 \\ -1 & 0 & 0 & 1 \\ 7 & -2 & 11 & -6 \\ 1 & 0 & -3 & 1 \end{bmatrix}.$$

Solution: Expanding with respect to the second column, and then with respect to the second row of the 3×3 matrix, we obtain

$$\begin{vmatrix} 1 & 0 & -1 & 2 \\ -1 & 0 & 0 & 1 \\ 7 & -2 & 11 & -6 \\ 1 & 0 & -3 & 1 \end{vmatrix} = -(-2) \begin{vmatrix} 1 & -1 & 2 \\ -1 & 0 & 1 \\ 1 & -3 & 1 \end{vmatrix} = 2 \left(-(-1) \begin{vmatrix} -1 & 2 \\ -3 & 1 \end{vmatrix} + (0) \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} - (1) \begin{vmatrix} 1 & -1 \\ 1 & -3 \end{vmatrix} \right)$$

$$= 2(5 + 0 - (-2)) = 14.$$

17. Consider the matrix $B = \begin{bmatrix} 3 & 2 & 1 \\ -2 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$.

(a) **[3 points]** By finding the roots of the characteristic polynomial, show that the eigenvalues of B are -1 and 2.

Solution: Starting with an expansion with respect to the third row, we have

$$\det(B-\lambda I) = \begin{vmatrix} 3-\lambda & 2 & 1 \\ -2 & -2-\lambda & -1 \\ 0 & 0 & -1-\lambda \end{vmatrix} = (-1-\lambda)\begin{vmatrix} 3-\lambda & 2 \\ -2 & -2-\lambda \end{vmatrix} = (-1-\lambda)(\lambda^2-\lambda-2) = -(\lambda+1)^2(\lambda-2).$$

Thus, the eigenvalues are -1 and 2.

(b) **[4 points]** For each of the eigenvalues of B found in part (a), find a basis of the corresponding eigenspace.

There is extra space for this question on the next page.

Solution: For the eigenvalue $\lambda = -1$, we have:

$$\left[\begin{array}{c|c} B-(-1)I & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 4 & 2 & 1 & 0 \\ -2 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + \frac{1}{2}R_1} \left[\begin{array}{ccc|c} 4 & 2 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 \to -2R_2} \left[\begin{array}{ccc|c} 4 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{4}R_1} \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Thus, the eigenspace is given by

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix}$$

and a basis of this eigenspace is

$$\left\{ \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 0 \end{bmatrix} \right\}.$$

For the eigenvalue $\lambda = 2$, we have:

$$\left[\begin{array}{c|c} B-2I & 0 \end{array}\right] = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ -2 & -4 & -1 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right]$$

$$\xrightarrow[R_1 \to R_1 - R_2]{R_3 \to R_3 + 3R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Thus, the eigenspace is given by

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad x_3 \in \mathbb{R},$$

and a basis of this eigenspace is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

(c) **[1 point]** Is B a diagonalizable matrix? Justify your answer.

Solution: B is not diagonalizable because there does not exist a basis consisting of the eigenvectors of B.

18. A European country has two private jet rental companies: *AeroNocia* and *GloboWings*. In September, $\frac{1}{4}$ of the country's businessmen flew with AeroNocia, while $\frac{3}{4}$ of the businessmen flew with GloboWings. After three months of tough marketing campaign between the two companies, $\frac{1}{4}$ of AeroNocia's customers switched to GloboWings, while $\frac{1}{2}$ of GloboWings' customers switched to AeroNocia.

(a) **[1 point]** Write down the migration matrix M and the initial state vector $\vec{x}_0$ for this problem.

Solution:

$$M = \begin{bmatrix} \frac{3}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}, \quad \vec{x}_0 = \begin{bmatrix} \frac{1}{4} \\ \frac{3}{4} \end{bmatrix}$$

Another possible migration matrix (if one reverses the order of the companies) is

$$M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{3}{4} \end{bmatrix} \quad \vec{x}_0 = \begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix}.$$

(b) **[1 point]** What is the market share of each of the companies at the end of the three-month marketing campaign?

Solution: $\vec{x}_1 = M\vec{x}_0 = \begin{bmatrix} \frac{9}{16} \\ \frac{7}{16} \end{bmatrix}$

(c) **[4 points]** Suppose that the same marketing campaign continues for several more three-month time periods, and its effect on customer migration remains the same. Then, in the long run, what is the predicted market share of each company? Put down your final answer in the following blank spots and write your detailed solution below.

In the long run, the market share of AeroNocia will be _____, and the market share of GloboWings will be _____.

Solution: To find the steady-state vector, we must find an eigenvector of eigenvalue 1. We row reduce:

$$\left[\begin{array}{c|c} M - I & 0 \end{array}\right] = \left[\begin{array}{cc|c} -\frac{1}{4} & \frac{1}{2} & 0 \\ \frac{1}{4} & -\frac{1}{2} & 0 \end{array}\right] \xrightarrow{R_2 \to R_2 + R_1} \left[\begin{array}{cc|c} -\frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -4R_1} \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

The general solution is thus

$$\vec{x} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

We choose the value of the free variable so that the sum of the entries of $\vec{x}$ is equal to one:

$$(2+1)\,x_2 = 1 \implies x_2 = \frac{1}{3}.$$

Thus the steady-state vector is

$$\vec{x} = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix}.$$

Since M is regular stochastic, the long term behaviour is given by the steady-state vector. Thus, in the long term, The maket share of AeroNocia will be $\frac{2}{3}$, and the market share of GloboWings will be $\frac{1}{3}$.

19. **[3 points]** Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & -4 & 2 & 3 \\ 1 & 1 & -2 & 0 & 8 \\ 2 & 0 & -8 & 5 & 5 \end{bmatrix}.$$

Find a basis for $\operatorname{Nul} A$.

Solution: We row reduce to find an echelon form of A.

$$\begin{bmatrix} 1 & 0 & -4 & 2 & 3 \\ 1 & 1 & -2 & 0 & 8 \\ 2 & 0 & -8 & 5 & 5 \end{bmatrix} \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - R_1} \begin{bmatrix} 1 & 0 & -4 & 2 & 3 \\ 0 & 1 & 2 & -2 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 2R_3} \begin{bmatrix} 1 & 0 & -4 & 0 & 5 \\ 0 & 1 & 2 & -2 & 5 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

$$\xrightarrow{R_2 \to R_2 + 2R_3} \begin{bmatrix} 1 & 0 & -4 & 0 & 5 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

Denoting the variables $x_1, \ldots, x_5$ as usual, we realize that x_1, x_2, x_4 are basic and x_3, x_5 are free. It follows that the general solution is

$$\begin{cases} x_1 = 4x_3 - 5x_5 \\ x_2 = -2x_3 - 3x_5 \\ x_4 = x_5 \end{cases}$$

and therefore the vector parametric form of the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -5 \\ -3 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

and the basis for $\operatorname{Nul} A$ is

$$\left\{ \begin{bmatrix} 4 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -3 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

20. [**4 points**] Let

$$A = \begin{bmatrix} 1 & -4 & -3 \\ 1 & -3 & -3 \\ -1 & 6 & 4 \end{bmatrix}$$

Find the inverse of A.

Solution: We row reduce the super-augmented matrix:

$$\left[\begin{array}{ccc|ccc} 1 & -4 & -3 & 1 & 0 & 0 \\ 1 & -3 & -3 & 0 & 1 & 0 \\ -1 & 6 & 4 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & -4 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & -4 & -3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -2 & 1 \end{array}\right] \xrightarrow{R_1 \to R_1 + 3R_3} \left[\begin{array}{ccc|ccc} 1 & -4 & 0 & 10 & -6 & 3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 + 4R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 6 & -2 & 3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -2 & 1 \end{array}\right]$$

Thus A is invertible, and its inverse is

$$A^{-1} = \begin{bmatrix} 6 & -2 & 3 \\ -1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}.$$

21. **[3 points]** Determine the dimension of $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 4 \end{bmatrix}, \quad \mathbf{v}_4 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}.$$

Solution: Note that $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is equal to $\text{Col}\, A$ where $A = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3\ \mathbf{v}_4]$. We now obtain an echelon form of A: We form the matrix whose columns are the given vectors and row reduce:

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ -1 & 0 & -1 & 2 \\ 1 & -1 & 0 & 1 \\ 2 & 2 & 4 & 3 \end{bmatrix} \xrightarrow[R_4 \to R_4 - 2R_1]{\substack{R_2 \to R_2 + R_1 \\ R_3 \to R_3 - R_1}} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & -2 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_4 \to R_4 - \frac{1}{6} R_3} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Clearly the pivot columns are columns 1, 2, and 4. Therefore the latter columns form a basis for $\text{Col}\, A$. It follows that the dimension of $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is equal to 3.